

SAIANKITH REDDY KOLLI

Amberg, Germany | ankithreddy580@gmail.com | +49 15510892792 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

M.Eng. student in AI for Smart Sensors and Actuators with hands-on project experience in machine learning, computer vision, smart sensors, and edge systems. Skilled in developing object-detection workflows, sensor-based prototypes, and interactive ML applications using Python, TensorFlow, Scikit-learn, and Streamlit.

ACADEMIC PROJECTS

Traffic Sign Detection for E-Bike Safety

Sep 2024 – Feb 2025

Deggendorf Institute of Technology Cham, Germany

- Tasked with building a reliable traffic sign detector for e-bike safety: custom-trained SSD MobileNetV2 320×320 FPN-Lite on a curated 20-class German traffic sign dataset (10,131 training / 365 validation images), achieving confidence scores above 97% and outperforming YOLOv11n on key sign classes.
- Responsible for the full SSD pipeline — dataset annotation via Roboflow, TFRecord conversion, augmentation, TensorFlow training over 50,000 steps — and delivered a published conference paper comparing SSD vs. YOLO using confusion matrices and confidence score analysis across all 20 classes.

Skills: TensorFlow, SSD MobileNetV2, YOLOv11n, Object Detection, OpenCV, Python

Decentralised UV Monitoring & Roller-Blind Control

Sep 2025 – Feb 2026

Deggendorf Institute of Technology Cham, Germany

- Tasked with hardware architecture and master-side control logic for a Raspberry Pi-based smart-home prototype: designed the UV sensing node (Joy-it SEN-UV01 + ADS1015 ADC via I2C) and master control unit (BLE communication, motor driver L298N, HMI display), successfully demonstrating UV data acquisition, CSV logging, and manual blind control.
- Addressed BLE library integration challenges through structured verification testing (5 test cases); documented partial system outcomes, identifying dependency conflicts as the root cause and outlining corrective steps, resulting in a complete technical report accepted at DIT.

Skills: Raspberry Pi, I2C, BLE, ADC Interfacing, MATLAB Simulink, Python, Embedded Systems

PERSONAL PROJECT

Drug Performance Analytics Suite

Aug 2025 – Sep 2025

drug-performance.streamlit.app

- Identified a gap in accessible pharma analytics tools: built an end-to-end Streamlit application analysing patient-reported drug feedback, integrating EDA dashboards, rating distributions, and condition-level visualisations to surface actionable insights from real-world drug review data.
- Engineered a medical-condition classification pipeline using Logistic Regression with Sentence Transformer embeddings and scaled numerical features, enabling real-time user-input predictions; deployed the app publicly on Streamlit Cloud for accessible, browser-based ML inference.

Skills: Streamlit, Scikit-learn, Logistic Regression, Sentence Transformers, Pandas, Python

EDUCATION

M.Eng. in AI for Smart Sensors and Actuators

Mar 2024 – Present

Deggendorf Institute of Technology Germany

B.Tech. in Chemical Engineering

Oct 2019 – May 2023

R.V.R. & J.C. College of Engineering India

WORK EXPERIENCE

Junior Engineer

Jun 2023 – Feb 2024

Hetero Labs Ltd. Hyderabad, India

- Tasked with process monitoring and documentation for industrial distillation systems: maintained Process Flow Diagrams (PFDs) and P&IDs, and supported real-time sensor-based monitoring of temperature, pressure, and flow parameters, ensuring compliance with operational standards.

- Gained hands-on exposure to sensors, actuators, and instrumentation in a regulated pharmaceutical manufacturing environment, directly informing later academic work in smart sensor systems.

Skills: *Process Engineering, P&ID Documentation, Sensors & Instrumentation, Technical Documentation*

Graduate Student Intern

Jun 2022 – Jul 2022

Aurobindo Pharma Ltd. Hyderabad, India

- Studied sensor and actuator applications for monitoring critical process parameters (temperature, pressure, flow) in pharmaceutical manufacturing, building foundational knowledge of real-world embedded sensing environments.
- Observed and documented control loop behaviour in GMP-compliant production lines, strengthening understanding of closed-loop process control relevant to later smart-sensor research.

Skills: *Sensors & Actuators, Pharmaceutical Processes, Process Monitoring*

TECHNICAL SKILLS

Programming:	Python, Typescript
AI / ML:	TensorFlow, Scikit-learn, Logistic Regression, Sentence Transformers
Computer Vision:	OpenCV, SSD MobileNetV2, YOLOv11n, Object Detection
Data:	Pandas, NumPy, Matplotlib, Seaborn
Sensors & Edge Systems:	Raspberry Pi, I2C, BLE, ADC Interfacing
Tools:	Streamlit, Git, Linux, MATLAB Simulink, GitHub

LANGUAGES

German (A2)	- Good
English	- Proficient
Telugu	- Mother Tongue
Hindi	- Proficient